

Corporate Employee Network Analysis using API Data

1. Data Source and Network Choice

For this assignment, I am using the Random User Generator Web API (randomuser.me) to build a dummy global employee network. By asking the API for 1,000 fake user profiles in JSON format, I am building a simple network graph with two tables. The Nodes table represents 1,000 employees, and the Edges table represents 3,000 internal work projects between them.

2. Categorical Variable

The category for each node is the employee's Regional Branch (like US, FR, or GB). I get this directly from the nationality code provided by the API.

3. High-Level Plan for Loading Data

I will do this analysis using Python in a notebook:

- Data Pull: Call the API using the Python requests library to get the JSON data.
- Data Setup: Read the JSON data into two Pandas tables. The first table is the Node list (Employee ID, Name, and Regional Branch). The second table is the Edge list, which randomly pairs Employee IDs to show who works with whom.
- Graph Creation: Load these tables into a NetworkX graph. The Employee IDs become the nodes, the Branch becomes the node attribute, and the work pairs become the edges.
- Analysis and Export: Calculate the centrality scores using NetworkX, group the scores by branch using Pandas, and save the network as a .graphml file to look at the graph in Gephi.

4. Hypothetical Outcome Prediction

If I calculate and compare degree centrality across the different Regional Branches, I predict that some locations (like the US or GB) will have a much higher average score than others. This would mean that employees in these main branches work with a lot more people and act as central hubs for the company, while employees in other branches work more on their own.

5. Network Questions to Ask

Using Chapters 2 and 3 of Tsvetovat's Social Network Analysis for Startups as a guide, I will ask these questions to understand the company network and employee power:

- **Graph Theory and Structure (Chapter 2):**

- a. Network Components: Is the whole company connected, or are there small groups of employees who never work with anyone else?
- b. Shortest Path and Distance: What is the fastest chain of introductions needed to pass a message between two distant employees, and exactly how many steps does that take?
- c. Diameter and Breadth-First Search: What is the longest distance between the two most disconnected employees in the whole company? If there is a company-wide announcement, how well does it spread layer by layer?

- **Centrality and Power (Chapter 3):**

- a. Degree Centrality: Which employees, and which branches on average, have the most direct work connections?
- b. Betweenness Centrality: Which employees act as important bridges between different offices? If these people left the company, would the network break apart?
- c. Closeness Centrality: Which employee is in the best spot to share a message with the whole company in the shortest amount of time?
- d. Eigenvector Centrality and PageRank: Which employees have the most hidden power because they work directly with the company's biggest hubs, instead of just having a lot of random connections?